

PROPERTIES OF T^2 TEST:

Some important properties of T^2 Test
for $H_0: \mu = \mu_0$
 $H_1: \mu \neq \mu_0$

are as follow:

4/6 (PU-23)

1) Invariant Property: (It is not correct, see after 2nd property)

Statement:

The T^2 Test is invariant to the transformation of the form

$$\underline{z} = \underline{C}'\underline{x} + \underline{b}$$

where ' $\underline{C}$ ' is a non-singular matrix of constants and ' $\underline{b}$ ' is a vector of constants.

Proof:

as $\underline{z} = \underline{C}'\underline{x} + \underline{b}$

$$\text{cov}(\underline{z}) = \underline{S}_z = \text{cov}(\underline{C}'\underline{x})$$

$$\underline{S}_z = \underline{C}' \text{cov}(\underline{x}) (\underline{C}')'$$

$$= \underline{C}' \text{cov}(\underline{x}) \underline{C} \quad \because \text{cov}(\underline{x}) = \underline{S}$$

$$\underline{S}_z = \underline{C}' \underline{S} \underline{C} \quad \text{--- (1)}$$

Consider $\xi_0 = c' u_0 + b$ (Mean value)

Then;

$$[T^2]_z = n [z - \xi_0]' S_z^{-1} [z - \xi_0]$$

$$(T^2)_z = n [c' \underline{x} + b - c' u_0 - b]' S_z^{-1} [c' \underline{x} + b - c' u_0 - b]$$

$$= n [c' (\underline{x} - u_0)]' S_z^{-1} [c' (\underline{x} - u_0)]$$

$$= n (\underline{x} - u_0)' (c')' S_z^{-1} c' (\underline{x} - u_0)$$

$$= n (\underline{x} - u_0)' c S_z^{-1} c' (\underline{x} - u_0)$$

$$\because S_z^{-1} = c' S c'$$

$$= n (\underline{x} - u_0)' c (c' S c')^{-1} c' (\underline{x} - u_0)$$

$$= n (\underline{x} - u_0)' c c^{-1} S^{-1} (c')^{-1} c' (\underline{x} - u_0)$$

$$= n (\underline{x} - u_0)' I S^{-1} I (\underline{x} - u_0)$$

$$= n (\underline{x} - u_0)' S^{-1} (\underline{x} - u_0)$$

$$\Rightarrow [T^2]_z = [T^2]_x$$

4Q Q: no: 04 (pu-23)

$$n = 140 \quad \mu = \begin{bmatrix} -3 \\ 2 \\ 0 \\ 1 \end{bmatrix} \text{ and}$$

let us partition Σ as

$$\Sigma = \begin{bmatrix} 8 & 2 & 3 & 1 \\ 2 & 5 & -1 & 3 \\ \hline 3 & -1 & 6 & -2 \\ 1 & 3 & -2 & 7 \end{bmatrix} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix}$$

$$\Sigma_{11} = \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix} \quad \Sigma_{12} = \begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}$$

$$\Sigma_{21} = \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix} \quad \Sigma_{22} = \begin{bmatrix} 6 & -2 \\ -2 & 7 \end{bmatrix}$$

Now we consider

$$\Sigma^{-1} = \Sigma_{11}^{-1} - \Sigma_{11}^{-1} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} \Sigma_{11}^{-1}$$

$$\Sigma^{-1} = \begin{bmatrix} 0.13889 & -0.0556 \\ -0.0556 & 0.2222 \end{bmatrix}$$

$$\Sigma_{cc}^{-1} = \begin{bmatrix} 0.1842 & 0.0526 \\ 0.0526 & 0.1578 \end{bmatrix}$$

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$$\sum_{i=1}^{-1} \sum_{j=1}^{-1} \sum_{k=1}^{-1} \sum_{l=1}^{-1} = \begin{bmatrix} 0.13889 & -0.05556 \\ -0.05556 & 0.2222 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ -1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 0.1842 & 0.0526 \\ 0.0526 & 0.1576 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 0.4729 & -0.0242 \\ -0.3890 & 0.6110 \end{bmatrix} \begin{bmatrix} 0.6052 & -0.0264 \\ 0.3156 & 0.4208 \end{bmatrix}$$

$$\sum_{i=1}^{-1} \sum_{j=1}^{-1} \sum_{k=1}^{-1} \sum_{l=1}^{-1} = \begin{bmatrix} 0.2770 & -0.0242 \\ -0.0426 & 0.2674 \end{bmatrix}$$

The characteristic equation is

$$\left| \sum_{i=1}^{-1} \sum_{j=1}^{-1} \sum_{k=1}^{-1} \sum_{l=1}^{-1} - \lambda I \right| = 0$$

$$\left| \begin{bmatrix} 0.2770 & -0.0242 \\ -0.0426 & 0.2674 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 0.2770 - \lambda & -0.0242 \\ -0.0426 & 0.2674 - \lambda \end{vmatrix} = 0$$

$$(0.2770 - \lambda)(0.2674 - \lambda) - (-0.0426)(-0.0242) = 0$$

$$0.0741678 - 0.2770\lambda - 0.2674\lambda + \lambda^2 - 0.00104 = 0$$

$$\lambda^2 - 0.5444\lambda + 0.0731278 = 0$$

Roots are - $\lambda_1 = 0.30388$

$\lambda_2 = 0.2406$

Sum of eigen values = Sum of diagonal matrix

Now the canonical correlation is

$$\rho_{u_1, v_1} = \sqrt{\lambda_1} = \sqrt{0.30328} = 0.5507$$

$$\rho_{u_2, v_2} = \sqrt{\lambda_2} = \sqrt{0.2406} = 0.4905$$

Now the eigen vectors corresponding to $\lambda_1 = 0.30328$ is

$$\left[\begin{matrix} \Sigma_{11}^{-1} \Sigma_{12} & \Sigma_{11}^{-1} \Sigma_{21} \end{matrix} \right] \underline{x} = \lambda_1 \underline{x}$$

$$\begin{bmatrix} 0.277 - \lambda & -0.023 \\ -0.041 & 0.2674 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.30328 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

OR

$$\left[\Sigma - \lambda I \right] \underline{x} = 0$$

$$\begin{bmatrix} 0.277 - \lambda & -0.023 \\ -0.041 & 0.2674 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

— eq (A)

$$\begin{bmatrix} +0.277 - 0.30328 & -0.023 \\ -0.041 & 0.2674 - 0.30328 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -0.02628 & -0.023 \\ -0.041 & -0.03628 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-0.02628x_1 - 0.023x_2 = 0$$

$$0.02628x_1 + 0.023x_2 = 0 \quad \text{— eq (2)}$$

$$-0.0426x_1 - 0.0364x_2 = 0$$

$$0.0426x_1 + 0.0364x_2 = 0 \quad \text{--- eq (1)}$$

from eq (1)

$$0.02628x_1 + 0.023x_2 = 0$$

$$0.02628x_1 = -0.023x_2 = 0$$

$$0.023x_2 = -0.02628x_1$$

$$0.02$$

adding eq (1) and eq (2)

$$0.0268x_1 + 0.0242x_2 = 0$$

$$0.0426x_1 + 0.0364x_2 = 0$$

$$0.0694x_1 + 0.0606x_2 = 0$$

let $x_1 = 1$ then $x_2 = 0.8732$

$$x = \begin{bmatrix} 1 \\ 0.8732 \end{bmatrix} \Rightarrow e_1 = \begin{bmatrix} 0.7532 \\ 0.6577 \end{bmatrix}$$

Since α_1 is subject to the constraint

$$\alpha_1' \sum_{i=1}^n \alpha_i = 1$$

Take $\alpha_1' \sum_{i=1}^n \alpha_i$

$$= \begin{bmatrix} 0.7532 & 0.6577 \end{bmatrix} \begin{bmatrix} 8 & 2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 0.7532 \\ 0.6577 \end{bmatrix}$$

$$= \begin{bmatrix} 0.7532 & 0.6577 \end{bmatrix} \begin{bmatrix} 7.3410 \\ 4.7949 \end{bmatrix}$$

$$= 5.5292 + 3.1536$$

$$= 8.6828 \neq 1$$

Now updated ' α_1 ' is

$$\underline{\alpha}_1 = \begin{bmatrix} 0.7532 / \sqrt{8.6828} \\ 0.6577 / \sqrt{8.6828} \end{bmatrix} = \begin{bmatrix} 0.2556 \\ 0.2232 \end{bmatrix}$$

$$\text{Now } \beta_1 = \lambda^{-1/2} \sum_{i=2}^p \sum_{j=1}^q \alpha_{ij}$$

$$\beta_1 = (0.3038)^{-1/2} \begin{bmatrix} 0.1842 & 0.0526 \\ 0.0526 & 0.1578 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0.2556 \\ 0.2232 \end{bmatrix}$$

$$\beta_1 = (0.3038)^{-1/2} \begin{bmatrix} 0.6052 & -0.0264 \\ 0.3156 & 0.4208 \end{bmatrix} \begin{bmatrix} 0.2556 \\ 0.2232 \end{bmatrix}$$

$$\beta_1 = (0.3038)^{-1/2} \begin{bmatrix} 0.1488 \\ 0.1746 \end{bmatrix}$$

$$\beta_1 = 1.8143 \begin{bmatrix} 0.1488 \\ 0.1746 \end{bmatrix} = \begin{bmatrix} 0.2700 \\ 0.3168 \end{bmatrix}$$

Now the first pair of canonical variates is;

$$u_1 = \underline{\alpha}_1' x_1 = \begin{bmatrix} 0.7532 & 0.6577 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$u_1 = 0.7532x_1 + 0.6577x_2$$

$$v_1 = \beta_1' x_2 = \begin{bmatrix} 0.2700 & 0.3168 \end{bmatrix} \begin{bmatrix} x_3 \\ x_4 \end{bmatrix}$$

$$v_1 = 0.2700x_3 + 0.3168x_4$$

Hence the two variates which are maximizing the correlation between two sets of variables (x_1, x_2) and (x_3, x_4) are u_1 and v_1 and the maximum correlation is : 0.3168

①

CPU-22

40 (Q:#:01)

Solution:

Given that

$$X_1 = \begin{bmatrix} 2 & 12 \\ 4 & 10 \\ 3 & 8 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 5 & 7 \\ 3 & 9 \\ 4 & 5 \end{bmatrix}$$

$$\begin{array}{r} \Sigma X_1 X_2 \\ 24 \\ 40 \\ 24 \\ \hline 88 \end{array}$$

$$n_1 = 3, \quad n_2 = 3, \quad p = 2$$

For $X_1 = \begin{bmatrix} 2 & 12 \\ 4 & 10 \\ 3 & 8 \end{bmatrix}$

$$\bar{X}_1 = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix}, \quad \bar{x}_1 = \frac{\Sigma x_1}{n} = \frac{2+4+3}{3} \Rightarrow \frac{9}{3} = 3$$

$$\bar{x}_2 = \frac{12+10+8}{3} \Rightarrow \frac{30}{3} = 10$$

$$\bar{X}_1 = [3 \quad 10]'$$

also;

$$S_{11} = \begin{bmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{bmatrix}, \quad s_{11} = \frac{\Sigma (x_1 - \bar{x}_1)^2}{n-1}$$

$$s_{11} = \frac{1}{3-1} [(2-3)^2 + (4-3)^2 + (3-3)^2] \Rightarrow 1$$

$$s_{12} = \frac{1}{n-1} [\Sigma x_1 x_2 - (\Sigma x_1)(\Sigma x_2)] \Rightarrow \frac{1}{3-1} [88 - (9)(30)] \Rightarrow -9$$

②

$$S_{22} = \frac{\sum (X_2 - \bar{X}_2)^2}{n-1}$$

$$= \frac{1}{3-1} [(12-10)^2 + (10-10)^2 + (8-10)^2] \Rightarrow 4$$

$$S_{11} = \begin{bmatrix} 1 & -91 \\ -91 & 4 \end{bmatrix}$$

For 2nd X_1 : $X_1 = \begin{bmatrix} 5 & 7 \\ 3 & 9 \\ 4 & 5 \end{bmatrix}$

$$\bar{X}_1 = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix}, \bar{x}_1 = \frac{\sum X_1}{n} = \frac{5+3+4}{3} = \frac{12}{3} = 4$$

$$\bar{x}_2 = \frac{\sum X_2}{n} = \frac{7+9+5}{3} = \frac{21}{3} = 7$$

$$\bar{X}_1 = [4 \quad 7]$$

$$S_2 = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$$

where $S_{11} = \frac{1}{n-1} \sum (X_1 - \bar{X}_1)^2 \Rightarrow \frac{1}{3-1} [(5-4)^2 + (3-4)^2 + (4-4)^2]$

$$S_{11} = 1$$

$$S_{12} = \frac{1}{n-1} [\sum X_1 X_2 - (\sum X_1)(\sum X_2)] \Rightarrow \frac{1}{3-1} [22 - (12)(21)]$$

$$S_{12} = S_{21} = -85$$

$$S_{22} = \frac{1}{n-1} \sum (X_2 - \bar{X}_2)^2 \Rightarrow \frac{1}{3-1} [(7-7)^2 + (9-7)^2 + (5-7)^2]$$

$$S_{22} = 4$$

$$S_2 = \begin{bmatrix} 1 & -85 \\ -85 & 4 \end{bmatrix}$$

Pooled estimate is

$$S = \frac{(n_1-1)S_1 + (n_2-1)S_2}{n_1+n_2-2} \Rightarrow \frac{(3-1)\begin{bmatrix} 1 & -91 \\ -91 & 4 \end{bmatrix} + (3-1)\begin{bmatrix} 1 & -85 \\ -85 & 4 \end{bmatrix}}{3+3-2}$$

$$= \frac{1}{4} \left\{ 2 \begin{bmatrix} 1 & -91 \\ -91 & 4 \end{bmatrix} + 2 \begin{bmatrix} 1 & -85 \\ -85 & 4 \end{bmatrix} \right\}$$

$$= \frac{1}{4} \left\{ \begin{bmatrix} 2 & -182 \\ -182 & 8 \end{bmatrix} + \begin{bmatrix} 2 & -170 \\ -170 & 8 \end{bmatrix} \right\}$$

$$= \frac{1}{4} \begin{bmatrix} 4 & -352 \\ -352 & 16 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & -88 \\ -88 & 4 \end{bmatrix}$$

Now the discrimination function is

$$Y = (\bar{x}_1 - \bar{x}_2)' S^{-1} x$$

$$S = \begin{bmatrix} 1 & -88 \\ -88 & 4 \end{bmatrix} \Rightarrow S^{-1} = \begin{bmatrix} 5 \times 10^{-4} & -0.011 \\ -0.011 & 1 \times 10^{-4} \end{bmatrix}$$

$$S^{-1} = \begin{bmatrix} 5 \times 10^{-4} & -0.011 \\ -0.011 & 1 \times 10^{-4} \end{bmatrix}$$

$$\bar{x}_1 - \bar{x}_2 = \begin{bmatrix} 3 \\ 10 \end{bmatrix} - \begin{bmatrix} 4 \\ 7 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$Y = \begin{bmatrix} -1 & 3 \end{bmatrix} \begin{bmatrix} 5 \times 10^{-4} & -0.011 \\ -0.011 & 1 \times 10^{-4} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -0.0335 & 0.0107 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = Y = -0.0335x_1 + 0.0107x_2$$

Classification Rule:

$$\bar{y}_1 = (\bar{x}_1 - \bar{x}_2)' S^{-1} \bar{x}_1$$

$$y_1 = \begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} 5 \times 10^{-4} & -0.011 \\ -0.011 & -1 \times 10^{-4} \end{bmatrix} \begin{bmatrix} 3 \\ 10 \end{bmatrix}$$

$$y_1 = 6.5 \times 10^{-3}$$

$$y_2 = (\bar{x}_1 - \bar{x}_2)' S^{-1} \bar{x}_2$$

$$= \begin{bmatrix} 1 & 3 \end{bmatrix} \begin{bmatrix} 5 \times 10^{-4} & -0.011 \\ -0.011 & -1 \times 10^{-4} \end{bmatrix} \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

$$y_2 = -0.0591$$

$$\text{Midpoint} = \frac{1}{2} (\bar{y}_1 + \bar{y}_2) = \frac{1}{2} (6.5 \times 10^{-3} + (-0.0591))$$
$$\Rightarrow -0.02625$$

Allocate the new observation x to population p_1 if

$$w = y - \frac{1}{2} (\bar{y}_1 + \bar{y}_2) > 0$$

and to the population p_2 otherwise

$$w = y + 0.02625 > 0$$

For the new observation

$$x = \begin{bmatrix} 1 \\ 4.4 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 4.4 \end{bmatrix} \begin{matrix} -0.02625 \end{matrix}$$

$$w = -0.0335(1) + 0.0107(4.4) \Rightarrow -0.0335 + 0.04708 = 0.01358$$

Principal Component Method To Extract
Common Factor;
OR

Principal Factor Method:

Question: (PU-22) L/O Repeat 2020 (b)

Consider the eigen value - eigen vector pair associated with the sample correlation matrix, attributed by the responses on 7 point semantic differential scale on 5 variables.

Taste (X_1), Good buy for Money (X_2), Flavor (X_3), Suitable for snack (X_4) & provides lots of energy (X_5).

Carry out Common Factor Analysis.

$$\lambda_1 = 2.85 \quad ; \quad e_1 = [0.33 \quad 0.46 \quad 0.38 \quad 0.56 \quad 0.47]'$$

$$\hat{\lambda}_1 = 1.81 ; e_1 = [-0.61 \quad 0.39 \quad -0.56 \quad 0.68 \quad 0.47]'$$

$$\hat{\lambda}_2 = 0.2 ; e_2 = [-0.1 \quad -0.71 \quad -0.17 \quad 0.6 \quad 0.22]'$$

$$\hat{\lambda}_3 = 0.1 ; e_3 = [-0.14 \quad 0.28 \quad -0.12 \quad 0.57 \quad -0.75]'$$

$$\hat{\lambda}_4 = 0.03 ; e_4 = [-0.7 \quad -0.01 \quad 0.71 \quad 0 \quad -0.01]'$$

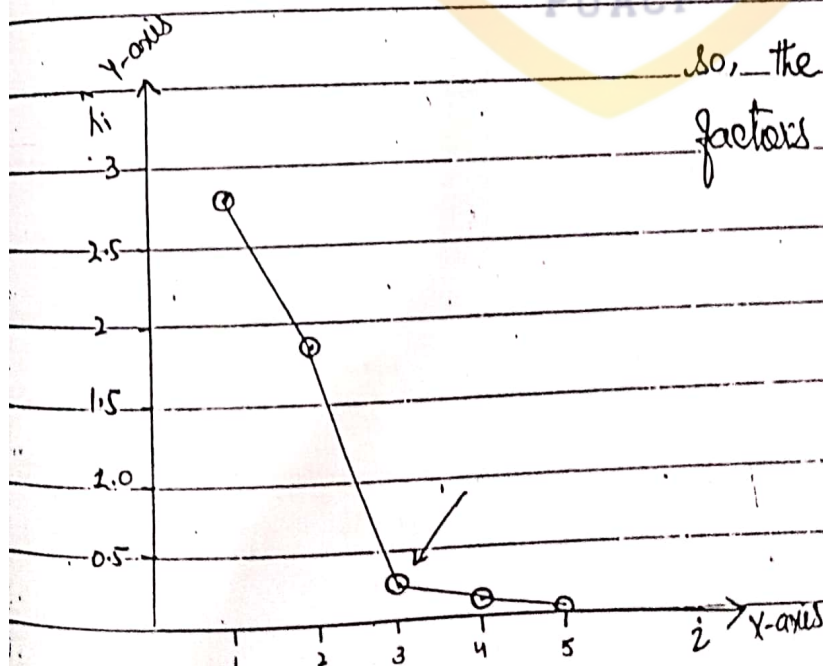
Solution:

(1) Decision regarding the No. of factors to be retained:

(a) KAISER CRITERIA:

As No. of eigen values greater than 1 are "2" i.e. $\hat{\lambda}_1 = 2.85$ & $\hat{\lambda}_2 = 1.81 > 1$
So, the No. of factors to be retained is $m = 2$

(b) SCREE Plot:



So, the No. of retained factors = $m = 2$

(e) Percentage of Explained Variation:

Cumulative %age of explained variation by

$$F_1 \quad E_p \quad F_2 = \frac{\hat{\lambda}_1 + \hat{\lambda}_2}{\hat{\lambda}_1 + \hat{\lambda}_2 + \hat{\lambda}_3 + \hat{\lambda}_4 + \hat{\lambda}_5} \times 100.$$

$$= \frac{2.85 + 1.81}{2.85 + 1.81 + 0.2 + 0.1 + 0.03} \times 100.$$

$$93\% > 50\%$$

So, the No. of factors which are to be retained is $m = 2$.

(2) Estimation:

Variable	Factor Loadings		Communalities	Specific Variance
	$F_1(\lambda_{1i})$	$F_2(\lambda_{2i})$	$\tilde{h}_i^2 = \lambda_{1i}^2 + \lambda_{2i}^2$	$\psi_i = \text{Var}(x_i) - \tilde{h}_i^2$
	$\sqrt{\lambda_1} e_1$	$\sqrt{\lambda_2} e_2$		
Taste (x_1)	0.55	-0.82	0.98	0.02
Good buy for Money (x_2)	0.77	0.53	0.89	0.11
Flavor (x_3)	0.64	-0.76	0.99	0.01
Suitable for Snack (x_4)	0.95	0.108	0.91	0.09
Provides lots of energy (x_5)	0.79	0.54	0.92	0.08
PEV:	57.11%	36.27%		

37. Factor Identification:

Factor - 1 (F_1) (Nutritional Factor)

X_2 : Good buy for Money

X_4 : Suitable for snack.

X_5 : Provides lots of energy.

Factor - 2 (F_2) (Taste Factor)

X_1 : Taste.

X_3 : Flavor.

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Question: 2010 E/B (PU-22) **Repeat -21**

Data for two variables give the summary. $n=42$ $\bar{x} = \begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix}$ Σ

$$\Sigma = \begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix}$$

- (a) Obtain 95% Confidence Ellipse for μ .
- (b) Determine the center, lengths & directions of major & minor axis of the ellipse.
- (c) Use the confidence ellipse to test the hypothesis $H_0: \mu = \begin{bmatrix} 0.562 \\ 0.589 \end{bmatrix}$. The above data is derived from $N_p(\mu, \Sigma)$.

SOLUTION:

(a) Confidence Region/Ellipse:

As,

$$S = \begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.046 \end{bmatrix}$$

$$S^{-1} = \begin{bmatrix} 199.0456 & -159.5092 \\ -159.5092 & 196.3190 \end{bmatrix}$$

also;

$$c^2 = \left(\frac{np}{n-p+1} \right) \frac{F(\alpha)}{(p, n-p+1)} \quad \because V = n-1$$

$$\therefore n = 42$$

$$c^2 = \frac{(42-1)(2)}{(42-1-2+1)} \frac{F_{0.05}}{(2, 42-2)} \quad \because p = 2$$

$$c^2 = (2.05)(3.23)$$

$$c^2 = 6.6215$$

The 95% Confidence Ellipse is:

$$n(\bar{x} - \mu)' S^{-1} (\bar{x} - \mu) \leq c^2$$

$$42 \begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix} - \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \begin{bmatrix} 199.04 & -159.509 \\ -159.509 & 196.319 \end{bmatrix} \begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix} - \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \leq 6.6215$$

$$42 \begin{bmatrix} 0.564 - \mu_1 & 0.603 - \mu_2 \\ 0.603 - \mu_2 & 0.564 - \mu_1 \end{bmatrix} \begin{bmatrix} 199.04 & -159.509 \\ -159.509 & 196.319 \end{bmatrix} \begin{bmatrix} 0.564 - \mu_1 \\ 0.603 - \mu_2 \end{bmatrix} = 0$$

(b) The center of the confidence ellipse will be at

$$\bar{x} = \begin{bmatrix} 0.564 \\ 0.603 \end{bmatrix}$$

Now for directions & lengths:

$$S = \begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix}$$

The characteristic equation is:

$$|S - \lambda I| = 0$$

$$\begin{vmatrix} 0.0144 - \lambda & 0.0117 \\ 0.0117 & 0.0146 - \lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 0.0144 - \lambda & 0.0117 \\ 0.0117 & 0.0146 - \lambda \end{vmatrix} = 0$$

$$(0.0144 - \lambda)(0.0146 - \lambda) - (0.0117)^2 = 0$$

$$0.00021024 - 0.0144\lambda - 0.0146\lambda + \lambda^2 - 0.00013689 = 0$$

$$\lambda^2 - 0.029\lambda + 0.00007335 = 0$$

$$a = 1, \quad b = -0.029, \quad c = 0.00007335$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda = \frac{-(-0.029) \pm \sqrt{(-0.029)^2 - 4(1)(0.00007335)}}{2(1)}$$

$$\lambda = \frac{0.029 \pm 0.0234}{2}$$

$$\lambda = \frac{0.029 + 0.0234}{2}, \quad \lambda = \frac{0.029 - 0.0234}{2}$$

$$\lambda_1 = 0.0262, \quad \lambda_2 = 0.0028$$

Now Eigen vectors:

$$S \underline{x} = \lambda_1 \underline{x}$$

$$\begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = (0.0262) \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\rightarrow 0.0144x_1 + 0.0117x_2 - 0.0262x_1 = 0$$

$$-0.0118x_1 + 0.0117x_2 = 0$$

$$0.0118x_1 - 0.0117x_2 = 0$$

$$\rightarrow 0.0117x_1 + 0.0146x_2 - 0.0262x_2 = 0$$

$$0.0117x_1 - 0.0116x_2 = 0$$

$$\text{let } x_1 = 1, \quad x_2 = 1$$

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \underline{e}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$$

Now for λ_2 :

$$S \underline{x} = \lambda_2 \underline{x}$$

$$\begin{bmatrix} 0.0144 & 0.0117 \\ 0.0117 & 0.0146 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.0028x_1 \\ 0.0028x_2 \end{bmatrix}$$

$$\rightarrow 0.0144x_1 + 0.0117x_2 - 0.0028x_1 = 0$$

$$0.0116x_1 + 0.0117x_2 = 0$$

$$\rightarrow 0.0117x_1 + 0.0146x_2 - 0.0028x_2 = 0$$

$$0.0117x_1 + 0.0118x_2 = 0$$

$$\text{let: } x_1 = 1, \quad x_2 = -1$$

$$\Rightarrow x = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \Rightarrow e_1 = \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix}$$

Length of Major Axis:

$$\Rightarrow 2 \sqrt{\frac{\lambda_1 c^2}{n}} = 2 \sqrt{\frac{(0.0262)(6.6215)}{42}} = 0.1285$$

Length of Minor Axis:

$$\Rightarrow 2 \sqrt{\frac{\lambda_2 c^2}{n}} = 2 \sqrt{\frac{(0.0028)(6.6215)}{42}} = 0.0420$$

Directions:

The major axis will lie in the direction of $e_1 = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$ and the minor axis will lie in the direction of $e_2 = \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix}$

(e) $T_0 = t$ $H_0:$

$$= \begin{bmatrix} 0.562 \\ 0.589 \end{bmatrix}$$

 $H_1:$

$$\neq \begin{bmatrix} 0.562 \\ 0.589 \end{bmatrix}$$

Substitute this value in eq (A), we get

$$(42) \begin{bmatrix} 0.002 & 0.014 \end{bmatrix} \begin{bmatrix} 199.04 & -159.509 \\ -159.509 & 196.319 \end{bmatrix} \begin{bmatrix} 0.002 \\ 0.014 \end{bmatrix} \leq 6.6215$$

$$- (42) \begin{bmatrix} 0.002 & 0.014 \end{bmatrix} \begin{bmatrix} -1.835 \\ 2.4294 \end{bmatrix} \leq 6.6215$$

$$(42) (0.00367 + 0.03034) \leq 6.6215$$

$$1.274 \leq 6.6215$$

Hence, $H_0: \underline{u_1} = \begin{bmatrix} 0.562 \\ 0.589 \end{bmatrix}$ will be

rejected. accepted

:- Past Paper 2015:-

Question No. 4:- 2021

Let X_1 and X_2 be two random variables with covariance matrix.

$$\Sigma = \begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix}$$

- i- Obtain the Principal Components and find the percentage of variation explained by each.
- ii- Change Σ into correlation matrix ρ . And find Principal Components using ρ .
- iii- Are Principal Components from (i) and (ii) same? If yes, why?

Solution:-

$$\Sigma = \begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix}$$

- i- Principal Components:-

The characteristic equation is:
 $|\Sigma - \lambda I| = 0$

$$\left| \begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 9-\lambda & \sqrt{6} \\ \sqrt{6} & 4-\lambda \end{bmatrix} \right| = 0$$

$$(9-\lambda)(4-\lambda) - (\sqrt{6})^2 = 0$$

$$36 - 9\lambda - 4\lambda + \lambda^2 - 6 = 0$$

$$\lambda^2 - 13\lambda + 30 = 0$$

$$\lambda = \frac{-(-13) \pm \sqrt{(-13)^2 - 4(1)(30)}}{2(1)}$$

$$\lambda = \frac{13 \pm \sqrt{169 - 120}}{2}; \lambda = \frac{13 \pm \sqrt{49}}{2}$$

$$\lambda = \frac{13 \pm 7}{2}$$

$$\lambda_1 = \frac{13+7}{2}; \lambda_2 = \frac{13-7}{2}$$

$$\boxed{\lambda_1 = 10}; \boxed{\lambda_2 = 3}$$

Eigen vector for $\lambda_1 = 10$

$$\sum X = \lambda_1 X$$

$$\begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 10 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$9x_1 + \sqrt{6}x_2 = 10x_1$$

$$9x_1 + \sqrt{6}x_2 - 10x_1 = 0$$

$$-x_1 + \sqrt{6}x_2 = 0 \quad \text{--- (i)}$$

$$\sqrt{6} x_1 + 4x_2 = 10x_2$$

$$\sqrt{6} x_1 + 4x_2 - 10x_2 = 0$$

$$\sqrt{6} x_1 - 6x_2 = 0 \longrightarrow (2)$$

let $x_1 = 1$ put in (1).

$$-1 + \sqrt{6} x_2 = 0$$

$$x_2 = 1/\sqrt{6}$$

$$\Rightarrow \boxed{x_2 = 0.4082}$$

$$x_1 = \begin{bmatrix} 1 \\ 0.4082 \end{bmatrix} \Rightarrow \underline{e_1} = \begin{bmatrix} \frac{1/\sqrt{1^2 + 0.4082^2}}{0.4082/\sqrt{1^2 + 0.4082^2}} \end{bmatrix}$$

$$\underline{e_1} = \begin{bmatrix} 0.9258 \\ 0.3779 \end{bmatrix}$$

Eigen vector for $\lambda_2 = 3$

$$\sum x = \lambda_2 x$$

$$\begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 3 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$9x_1 + \sqrt{6} x_2 = 3x_1$$

$$9x_1 - 3x_1 + \sqrt{6} x_2 = 0$$

$$6x_1 + \sqrt{6} x_2 = 0 \longrightarrow (3)$$

$$\sqrt{6} x_1 + 4x_2 = 3x_2$$

$$\sqrt{6} x_1 + 4x_2 - 3x_2 = 0$$

$$\sqrt{6} x_1 + x_2 = 0 \longrightarrow (4)$$

let $x_1 = 1$ in (3)

$$6 + \sqrt{6} x_2 = 0$$

$$x_2 = \frac{-6}{\sqrt{6}} = -2.4495$$

$$X = \begin{bmatrix} 1 \\ -2.4495 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} \frac{1/\sqrt{1^2 + (-2.4495)^2}}{-2.4495/\sqrt{1^2 + (-2.4495)^2}} \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 0.3780 \\ -0.9258 \end{bmatrix}$$

Principal Components:-

$$1^{st} PC = Y_1 = \underline{e}_1' X = [0.9258 \quad 0.3779] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$Y_1 = 0.9258x_1 + 0.3779x_2$$

$$2^{nd} PC = Y_2 = \underline{e}_2' X = [0.3780 \quad -0.9258] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$Y_2 = 0.3780x_1 - 0.9258x_2$$

Percentage of variation explained:-

%age of explained variation by PC-I:

$$= \frac{\lambda_1}{\lambda_1 + \lambda_2} \times 100$$

$$= \frac{10}{10+3} \times 100 = 76.92\%$$

%age of explained variation by PC-II:-

$$= \frac{\lambda_2}{\lambda_1 + \lambda_2} \times 100 = \frac{3}{10+3} \times 100$$

$$= 23.077\%$$

$$\alpha' \Sigma_{11} \alpha = 1$$

and then this is used to obtain the value of β using eq. (4).

Question: (2012)

V10
(94-20)

The covariance matrix for the four standardized variables z_1, z_2, z_3 & z_4 is:

$$\rho = \begin{bmatrix} 1 & 0.4 & 0.5 & 0.6 \\ 0.4 & 1 & 0.3 & 0.4 \\ 0.5 & 0.3 & 1 & 0.2 \\ 0.6 & 0.4 & 0.2 & 1 \end{bmatrix}$$

Considering $z_1 = [z_1, z_2]'$ & $z_2 = [z_3, z_4]'$.
Obtain Canonical correlation between z_1 & z_2 . Also find the first pair of Canonical variates.

Solution:

Given that

$$\rho = \begin{bmatrix} 1 & 0.4 & 0.5 & 0.6 \\ 0.4 & 1 & 0.3 & 0.4 \\ 0.5 & 0.3 & 1 & 0.2 \\ 0.6 & 0.4 & 0.2 & 1 \end{bmatrix} = \begin{bmatrix} \rho_{11} & \rho_{12} \\ \rho_{21} & \rho_{22} \end{bmatrix}$$

Now Consider;

$$L_{11}^{-1} = \begin{bmatrix} 1 & 0.4 \\ 0.4 & 1 \end{bmatrix}, \quad L_{12}^{-1} = \begin{bmatrix} 0.5 & 0.6 \\ 0.3 & 0.4 \end{bmatrix}$$

$$L_{21}^{-1} = \begin{bmatrix} 0.5 & 0.3 \\ 0.6 & 0.4 \end{bmatrix}, \quad L_{22}^{-1} = \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix}$$

$$L_{11}^{-1} = \begin{bmatrix} 1 & 0.4 \\ 0.4 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1.1905 & -0.4762 \\ -0.4762 & 1.1905 \end{bmatrix}$$

$$L_{22}^{-1} = \begin{bmatrix} 1 & 0.2 \\ 0.2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1.0416 & -0.2083 \\ -0.2083 & 1.0416 \end{bmatrix}$$

$$\Rightarrow L_{11}^{-1} L_{12}^{-1} L_{22}^{-1} L_{21}^{-1} = \begin{bmatrix} 1.1905 & -0.4762 \\ -0.4762 & 1.1905 \end{bmatrix} \begin{bmatrix} 0.5 & 0.6 \\ 0.3 & 0.4 \end{bmatrix} \begin{bmatrix} 1.0416 & -0.2083 \\ -0.2083 & 1.0416 \end{bmatrix}$$

$$L_{11}^{-1} L_{12}^{-1} L_{22}^{-1} L_{21}^{-1} = \begin{bmatrix} 0.45239 & 0.52382 \\ 0.11905 & 0.1908 \end{bmatrix} \begin{bmatrix} 0.39582 & 0.22916 \\ 0.52081 & 0.35415 \end{bmatrix}$$

$$L_{11}^{-1} L_{12}^{-1} L_{22}^{-1} L_{21}^{-1} = \begin{bmatrix} 0.4519 & 0.2892 \\ 0.1463 & 0.0947 \end{bmatrix}$$

The characteristic eq is

$$|L_{11}^{-1} L_{12}^{-1} L_{22}^{-1} L_{21}^{-1} - \lambda I| = 0.$$

$$\begin{vmatrix} 0.4519 & 0.2892 \\ 0.1463 & 0.0947 \end{vmatrix} - \lambda \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 0.$$

$$\begin{vmatrix} 0.4519 & 0.2892 \\ 0.1463 & 0.0947 \end{vmatrix} - \begin{vmatrix} \lambda & 0 \\ 0 & \lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 0.4519 - \lambda & 0.2892 \\ 0.1463 & 0.0947 - \lambda \end{vmatrix} = 0$$

$$(0.4519 - \lambda)(0.0947 - \lambda) - 0.0423 = 0$$

$$0.0428 - 0.4519\lambda - 0.0947\lambda + \lambda^2 - 0.0423 = 0$$

$$\lambda^2 - 0.5466\lambda + 0.0005 = 0$$

$$\Rightarrow \lambda_1 = 0.5456, \quad \lambda_2 = 0.0009163.$$

So, the Canonical correlations are:

$$\rho_{u_1x} = \sqrt{\lambda_1} = \sqrt{0.5456} = 0.7386.$$

$$\rho_{u_2x} = \sqrt{\lambda_2} = \sqrt{0.0009163} = 0.03027.$$

Now, the eigen vector corresponding to $\lambda_1 = 0.5456$ is:

$$\begin{bmatrix} e_{11}^{-1} & e_{12}^{-1} & e_{22}^{-1} & e_{21}^{-1} \end{bmatrix} x = \lambda_1 x$$

$$\begin{bmatrix} 0.4519 & 0.2892 \\ 0.1463 & 0.0947 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.5456 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$0.4519x_1 + 0.2892x_2 = 0.5456x_1$$

$$0.4519x_1 - 0.5456x_2 + 0.2892x_2 = 0$$

$$-0.0937x_1 + 0.2892x_2 = 0$$

$$0.0937x_1 - 0.2892x_2 = 0 \rightarrow \textcircled{1}$$

$$0.1463x_1 + 0.0947x_2 - 0.5456x_2 = 0$$

$$0.1463x_1 - 0.4509x_2 = 0 \rightarrow (2)$$

Let $x_1 = 1$ then $x_2 = 0.3245$

$$x = \begin{bmatrix} 1 \\ 0.3245 \end{bmatrix} \Rightarrow e_1 = \begin{bmatrix} 0.9512 \\ 0.3087 \end{bmatrix} = \alpha_1 \text{ (Normalized eigen vector)}$$

Since, ' α ' is subject to the constraint:

$$\alpha' \Sigma \alpha = 1$$

Take, $\alpha' \Sigma \alpha$

$$= \begin{bmatrix} 0.9512 & 0.3087 \end{bmatrix} \begin{bmatrix} 1 & 0.4 \\ 0.4 & 1 \end{bmatrix} \begin{bmatrix} 0.9512 \\ 0.3087 \end{bmatrix}$$

$$= \begin{bmatrix} 0.9512 & 0.3087 \end{bmatrix} \begin{bmatrix} 0.9512 + 0.12348 \\ 0.38048 + 0.3087 \end{bmatrix}$$

$$= \begin{bmatrix} 0.9512 & 0.3087 \end{bmatrix} \begin{bmatrix} 1.07468 \\ 0.68918 \end{bmatrix}$$

$$= 1.0222 + 0.2127$$

$$= 1.2349 \neq 1$$

Now, updated ' α ' is:

$$\alpha_1 = \begin{bmatrix} 0.9512 / \sqrt{1.2349} \\ 0.3087 / \sqrt{1.2349} \end{bmatrix} = \begin{bmatrix} 0.8559 \\ 0.2778 \end{bmatrix}$$

Now;

$$B_1 = \lambda_1^{-1/2} P^{-1} P \alpha$$

$$B_1 = (0.5456)^{-1/2} \begin{bmatrix} 1.0416 & -0.2083 \\ -0.2083 & 1.0416 \end{bmatrix} \begin{bmatrix} 0.5 & 0.3 \\ 0.6 & 0.4 \end{bmatrix} \begin{bmatrix} 0.8559 \\ 0.2778 \end{bmatrix}$$

$$B_1 = (0.5456)^{-1/2} \begin{bmatrix} 1.0416 & -0.2083 \\ -0.2083 & 1.0416 \end{bmatrix} \begin{bmatrix} 0.51129 \\ 0.62466 \end{bmatrix}$$

$$B_1 = (0.5456)^{-1/2} \begin{bmatrix} 0.4024 \\ 0.5441 \end{bmatrix}$$

$$B_1 = \begin{bmatrix} 0.5447 \\ 0.73661 \end{bmatrix}$$

Now; The first pair of canonical variables is

$$u_1 = \alpha'_1 z_1 = \begin{bmatrix} 0.8559 & 0.2778 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$u_1 = 0.8559 z_1 + 0.2778 z_2$$

$$v_1 = \beta'_1 z_2 = \begin{bmatrix} 0.5447 & 0.73661 \end{bmatrix} \begin{bmatrix} z_3 \\ z_4 \end{bmatrix}$$

$$v_1 = 0.5447 z_3 + 0.73661 z_4$$

Here, the two variables which are maximizing the correlation between two sets of variables (z_1, z_2) & (z_3, z_4) are 'u₁' and 'v₁' and the maximum correlation is: 0.7386.

Question: 4/Q (Past Paper) (pu-20)

Consider the following matrix

$$\Sigma = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- 1) Define Principal Components.
- 2) Identify % age of explained variation by each PC.
- 3) Comment on dimension reduction (data reduction).

Solution:

The characteristic eq. is:

$$|\Sigma - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & -2 & 0 \\ 0 & 5-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 1-\lambda & -2 & 0 \\ -2 & 5-\lambda & 0 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0$$

By expanding From R_3 ; we have

$$(2-\lambda) \begin{vmatrix} 1-\lambda & -2 \\ -2 & 5-\lambda \end{vmatrix} = 0$$

$$2 - \lambda = 0 \Rightarrow \boxed{\lambda = 2} \text{ OR}$$

$$(1 - \lambda)(5 - \lambda) - 4 = 0$$

$$5 - 5\lambda - \lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 6\lambda + 1 = 0$$

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\lambda = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(1)}}{2(1)}$$

$$\lambda = \frac{6 \pm 5.6568}{2}$$

$$\lambda = 0.17, \quad \lambda = 5.83$$

So, the eigen values are:

$$\lambda_1 = 5.83, \quad \lambda_2 = 2, \quad \lambda_3 = 0.17$$

For eigen vectors:

i) For $\lambda_1 = 5.83$:

$$\sum x = \lambda_1 x$$

$$\begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 5.83x_1 \\ 5.83x_2 \\ 5.83x_3 \end{bmatrix}$$

$$x_1 - 2x_2 - 5.83x_1 = 0$$

$$-4.83x_1 - 2x_2 = 0$$

$$\Rightarrow 2.415x_1 + x_2 = 0 \quad \text{--- (1)}$$

$$-2x_1 + 5x_2 - 5.83x_2 = 0$$

$$-2x_1 + (-0.83x_2) = 0$$

$$\Rightarrow 2x_1 + 0.83x_2 = 0 \quad \text{--- (2)}$$

$$2x_3 - 0.83x_3 = 0 \Rightarrow \boxed{x_3 = 0}$$

$$\text{let } x_1 = 1 \Rightarrow x_2 = -2.415$$

$$x_1 = \begin{bmatrix} 1 \\ -2.415 \\ 0 \end{bmatrix} \Rightarrow e_1 = \begin{bmatrix} 0.3825 \\ -0.9239 \\ 0 \end{bmatrix}$$

ii) For $\lambda_2 = 2i$

$$\Sigma x = \lambda_2 x$$

$$\begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 2x_2 \\ -2x_3 \end{bmatrix}$$

$$x_1 - 2x_2 - 2x_3 = 0$$

$$\boxed{x_1 \neq 0} \quad -x_1 - 2x_2 = 0 \Rightarrow x_1 + 2x_2 = 0 \quad \text{--- (3)}$$

$$-2x_1 + 5x_2 - 2x_3 = 0$$

$$-2x_1 + 3x_2 = 0 \Rightarrow 2x_1 - 3x_2 = 0 \quad \text{--- (4)}$$

$$2x_3 = 2x_2$$

$$\text{let } x_3 = 1, \quad x_1 = 0, \quad x_2 = 0$$

$$x_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow e_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

iii) For $\lambda_3 = 0.17i$

$$\Sigma x = \lambda_3 x$$

$$\begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.17x_1 \\ 0.17x_2 \\ 0.17x_3 \end{bmatrix}$$

$$x_1 - 2x_2 - 0.17x_1 = 0$$

$$0.83x_1 - 2x_2 = 0 \quad \text{--- (5)}$$

$$-2x_1 + 5x_2 - 0.17x_2 = 0$$

$$-2x_1 + 4.83x_2 = 0$$

$$\Rightarrow x_1 - 2.415x_2 = 0 \quad \text{--- (6)}$$

$$2x_3 - 0.17x_3 = 0 \Rightarrow \boxed{x_3 = 0}$$

$$\text{Let } x_1 = 2.415, \quad x_2 = 1$$

$$x_3 = \begin{bmatrix} 2.415 \\ 1 \\ 0 \end{bmatrix} \Rightarrow e_3 = \begin{bmatrix} 0.9238 \\ 0.3825 \\ 0 \end{bmatrix}$$

a) Principal Components:

$$\text{1st PC} = Y_1 = e_1' X = \begin{bmatrix} 0.3825 & -0.9238 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$Y_1 = 0.3825x_1 - 0.9238x_2$$

$$\text{2nd PC} = Y_2 = e_2' X = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$Y_2 = x_3$$

$$\text{3rd PC} = Y_3 = e_3' X = \begin{bmatrix} 0.9238 & 0.3825 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$Y_3 = 0.9238x_1 + 0.3825x_2$$

Interpretation:

1st PC is a contrast between x_1 & x_2

and PC can exactly be reproduced by

1st and 3rd PC is a weighted

sum of x_1 & x_2

Percentage of explained variation:

% age of explained variation by 1st PC:

$$= \frac{\text{Var}(Y_1)}{\sum_{i=1}^n \text{Var}(Y_i)} \times 100$$

$$= \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} \times 100$$

$$= \frac{5.83}{5.83 + 2 + 0.17} \times 100 = 72.875\%$$

$$= \frac{5.83}{5.83 + 2 + 0.17} \times 100 = 72.875\%$$

% age of explained variation by 2nd PC:

$$= \frac{\lambda_2}{\lambda_1 + \lambda_2 + \lambda_3} \times 100$$

$$= \frac{2}{5.83 + 2 + 0.17} \times 100 = 25\%$$

$$= \frac{2}{5.83 + 2 + 0.17} \times 100 = 25\%$$

$$= \frac{2}{5.83 + 2 + 0.17} \times 100 = 25\%$$

% age of explained variation by 3rd PC:

$$= \frac{\lambda_3}{\lambda_1 + \lambda_2 + \lambda_3} \times 100 =$$

$$= \frac{0.17}{5.83 + 2 + 0.17} \times 100 = 2.125\%$$

$$= \frac{0.17}{8} \times 100 = 2.125\%$$

2) Comment on dimension reduction:

Keeping in view the explained variation, first two Principal Components Y_1, Y_2 can replace by the original variables X_1, X_2 & X_3 as this requires about 2% loss of variation.

Conclusion .

Date: _____

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Q1 (pu-20)

Solution:

Given that

$$n = 7$$

X_1

X_2

$$\sum X_1 X_2 = 9155$$

12

41

11

66

22

76

18

109

19

108

16

76

15

69

$$\sum X_1 = 113$$

$$\sum X_2 = 545$$

$$\bar{X}_1 = \frac{113}{7}$$

$$\bar{X}_2 = \frac{545}{7}$$

$$\boxed{\bar{X}_1 = 16.14}$$

$$\boxed{\bar{X}_2 = 77.8571}$$

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Confidence Region

As

$$S = \begin{bmatrix} s_{11} & s_{12} \\ s_{21} & s_{22} \end{bmatrix}$$

$$s_{11} = \frac{1}{n-1} \sum (x_i - \bar{x})^2 \text{ or}$$

$$= \frac{1}{7-1} \left[(12-16.14)^2 + (11-16.14)^2 + (22-16.14)^2 \right. \\ \left. + (18-16.14)^2 + (19-16.14)^2 + (16-16.14)^2 \right. \\ \left. + (15-16.14)^2 \right]$$

$$= \frac{1}{6} \left[17.1396 + 26.4196 + 34.3396 \right. \\ \left. + 3.4596 + 8.1796 + 0.0196 \right. \\ \left. + 1.2996 \right]$$

$$s_{11} = 15.14$$

$$s_{12} = \frac{1}{n-1} \left[\sum x_1 x_2 - (\sum x_1)(\sum x_2) \right]$$

$$= \frac{1}{7-1} [9155 - (113)(845)]$$

$$s_{12} = s_{21} = -8738.3$$

Date: _____

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$$S_{22} = \frac{1}{n-1} \sum (x_2 - \bar{x}_2)^2$$

$$= \frac{1}{7-1} [(41-77.85)^2 + (66-77.85)^2 + (76-77.85)^2 + (109-77.85)^2 + (108-77.85)^2 + (76-77.85)^2 + (69-77.85)^2]$$

$$= \frac{1}{6} [1357.9 + 140.4 + 3.42 + 970.32 + 909.02 + 3.42 + 78.3]$$

$$S_{22} = 577.13$$

$$S = \begin{pmatrix} 18.14 & -8738.3 \\ -8738.3 & 577.13 \end{pmatrix}$$

$$S^{-1} = \begin{pmatrix} -7 \times 10^{-6} & -1 \times 10^{-4} \\ -1 \times 10^{-4} & -1 \times 10^{-7} \end{pmatrix}$$

also;

$$C^2 = \left(\frac{V P}{V - P + 1} \right) F(\alpha) (P, V - P + 1)$$

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$$v = n - 1 = 7 - 1 = 6$$

$$n = 7$$

$$p = 2$$

$$C^2 = \left(\frac{vP}{v - P + 1} \right) F(\alpha)$$

$$C^2 = \left(\frac{(6 \times 2)}{6 - 2 + 1} \right) F(0.05)$$

$$(2, 6 - 2 + 1)$$

$$C^2 = \left(\frac{12}{5} \right) F_{(0.05)}$$

$$(2, 5)$$

$$C^2 = \left(\frac{12}{5} \right) (5.79)$$

$$C^2 = (2.4) (5.79)$$

$$C^2 = 13.896$$

The 95% Confidence

$$n(\bar{x} - \mu)' S^{-1} (\bar{x} - \mu) \leq C^2$$

$$\leq 13.896$$

$$7 \left(\begin{bmatrix} 16.14 \\ 77.85 \end{bmatrix} - \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \right)' \begin{bmatrix} -7 \times 10^{-6} & -1 \times 10^{-4} \\ -1 \times 10^{-4} & -1 \times 10^{-7} \end{bmatrix} \begin{bmatrix} 16.14 \\ 77.85 \end{bmatrix} - \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$$

$$7 \left[16.14\mu_1, 77.85 - \mu_2 \right] \begin{bmatrix} -7 \times 10^{-6} & -1 \times 10^{-4} \\ -1 \times 10^{-4} & -1 \times 10^{-7} \end{bmatrix} \begin{bmatrix} 16.14 - \mu_1 \\ 77.85 - \mu_2 \end{bmatrix} \leq 13.896$$

eq (A)

Date: _____

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The center of the confidence
will be at

$$\bar{x} = \begin{pmatrix} 15.14 \\ 77.85 \end{pmatrix}$$

Now for lengths:

$$S = \begin{pmatrix} 15.14 & -8738.3 \\ -8738.3 & 577.13 \end{pmatrix}$$

The characteristic equation is

$$|S - \lambda I| = 0$$

$$\left| \begin{pmatrix} 15.14 & -8738.3 \\ -8738.3 & 577.13 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = 0$$

$$\left| \begin{pmatrix} 15.14 - \lambda & -8738.3 \\ -8738.3 & 577.13 - \lambda \end{pmatrix} \right| = 0$$

$$(15.14 - \lambda)(577.13 - \lambda) - (-8738.3)(-8738.3) = 0$$

$$8737.7 - 15.14\lambda - 577.13\lambda + \lambda^2 - 76357886.89 = 0$$

$$\lambda^2 - 592.27\lambda - 76349149.19 = 0$$

$$\text{roots } \lambda_1 = 9038.95$$

$$\lambda_2 = -8446.68$$

Sum of eigen values = sum of diagonal elements

$$592.27 = 592.27$$

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Now eigen vectors

$$\left[\begin{pmatrix} A \\ B \end{pmatrix} - \lambda I \right] x = 0$$

$$\begin{bmatrix} 15.14 - \lambda & -8738.3 \\ -8738.3 & 577.13 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{--- eq (A)}$$

When $\lambda_1 = 9038.95$

$$\begin{bmatrix} 15.14 - 9038.95 & -8738.3 \\ -8738.3 & 577.13 - 9038.95 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -9023.81 & -8738.3 \\ -8738.3 & -8461.82 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-9023.81 x_1 - 8738.3 x_2 = 0$$

$$-8738.3 - 8461.82 x_2 = 0$$

$$9023.81 x_1 + 8738.3 x_2 = 0 \quad \text{--- eq (i)}$$

$$+8738.3 + 8461.82 x_2 = 0 \quad \text{--- eq (ii)}$$

0 0 0

from eq (1)

$$9023.81x_1 + 8738.3x_2 = 0$$

$$9023.81x_1 = -8738.3x_2$$

$$\frac{x_1}{8738.3} = \frac{-x_2}{9023.81}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 8738.3 \\ -9023.81 \end{bmatrix}$$

Normalized unit vector of $\underline{a} = \frac{1}{\|\underline{a}\|} \underline{a}$

$$\|\underline{a}\| = \sqrt{(8738.3)^2 + (-9023.81)^2}$$

$$= 12561.3$$

$$\underline{e}_1 = \frac{1}{12561.3} \begin{bmatrix} 8738.3 \\ -9023.81 \end{bmatrix} = \begin{bmatrix} \frac{8738.3}{12561.3} \\ \frac{-9023.81}{12561.3} \end{bmatrix}$$

$$\underline{e}_1 = \begin{bmatrix} 0.695 \\ -0.71 \end{bmatrix}$$

When $\lambda = -8446.68$

$$\begin{bmatrix} S - \lambda I \\ u \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 15.14 - \lambda & -8738.3 \\ -8738.3 & 577.13 - \lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 15.14 - (-8446.68) & -8738.3 \\ -8738.3 & 577.13 - (-8446.68) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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$$\begin{pmatrix} 8461.82 & -8738.3 \\ -8738.3 & 9023.81 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$8461.82x_1 - 8738.3x_2 = 0 \quad \text{--- eq (i)}$$

$$-8738.3x_1 - 9023.81x_2 = 0 \quad \text{--- eq (ii)}$$

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from eq (iii)

$$8461.82x_1 - 8738.3x_2 = 0$$

$$8461.82x_1 = 8738.3x_2$$

$$\frac{x_1}{8738.3} = \frac{x_2}{8461.82}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 8738.3 \\ 8461.82 \end{bmatrix}$$

Normalized unit vector of $a = \frac{1}{\|a\|} a$

$$\|a\| = \sqrt{(8738.3)^2 + (8461.82)^2}$$

$$= 12163.89$$

$$e = (12163.89)^{-1} \begin{bmatrix} 8738.3 \\ 8461.82 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{8738.3}{12163.89} \\ \frac{8461.82}{12163.89} \end{bmatrix} \Rightarrow \begin{bmatrix} 0.71 \\ 0.69 \end{bmatrix}$$

length of major axis

$$2\sqrt{\frac{A_1 C^2}{n}} \Rightarrow 2\sqrt{\frac{(9038.25)(13.896)}{7}} \Rightarrow 267.9$$

length minor axis

$$2\sqrt{\frac{A_2 C^2}{n}} \Rightarrow 2\sqrt{\frac{(-8446.68)(13.896)}{7}} \Rightarrow 2\sqrt{\frac{11735.02}{7}}$$